

第八章 力学变分原理

§8.1 变分法初步

1. 泛函

有一光滑轨道 $y = y(x)$

质点沿轨道下滑时间: $J = \int dt = \int \frac{ds}{v} = \int \frac{\sqrt{1+y'^2}}{\sqrt{2gy}} dx$

此时 J 取决于函数 $y = y(x)$, 则称 J 是 $y(x)$ 的泛函, 记作 $J[y(x)]$

2 变分问题

求泛函 $J[y(x)]$ 的极值.

3. 欧拉方程

$$\text{泛函 } J[y(x)] = \int_a^b F(x, y, y') dx$$

函数 $y = y(x)$ 的变分: $y \rightarrow y + \delta y$

$$\text{相应地: } \delta J = J[y + \delta y] - J[y]$$

$$= \int_a^b F(x, y + \delta y, y' + \delta y') dx - \int_a^b F(x, y, y') dx$$

$$\approx \int_a^b \left(\frac{\partial F}{\partial y} \cdot \delta y + \frac{\partial F}{\partial y'} \cdot \delta y' \right) dx$$

假设 $y = y(x)$ 使 $J[y(x)]$ 取得极值, 记 $\delta y = \varepsilon \eta$, ε 为小参数

$$\text{此时 } \delta y = \varepsilon \eta, \delta y' = \varepsilon \eta'$$

$$\text{极值条件: } \left. \frac{\partial J(y + \varepsilon \eta(x))}{\partial \varepsilon} \right|_{\varepsilon=0} = 0$$

$$\text{即 } \frac{\partial J}{\partial y} \cdot \eta + \frac{\partial J}{\partial y'} \cdot \eta' = 0 \rightarrow \int_a^b \left(\frac{\partial J}{\partial y} \cdot \eta + \frac{\partial J}{\partial y'} \cdot \eta' \right) dx = 0$$

差一个乘数

①

$$\delta J[y] = \int_a^b \left(\frac{\partial F}{\partial y} \cdot \delta y + \frac{\partial F}{\partial y'} \cdot \delta y' \right) dx = 0$$

分部积分法 $\rightarrow$ 消去 $\delta y'$

$$\begin{aligned} \delta J[y] &= \int_a^b \left(\frac{\partial F}{\partial y} \cdot \delta y + \frac{\partial F}{\partial y'} \cdot \delta y' \right) dx \\ &= \int_a^b \frac{\partial F}{\partial y} \cdot \delta y \, dx + \left. \frac{\partial F}{\partial y'} \cdot \delta y \right|_a^b - \int_a^b \left[\frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \cdot \delta y \right] dx \\ &= \int_a^b \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] \delta y \, dx + \left. \frac{\partial F}{\partial y'} \cdot \delta y \right|_a^b = 0 \end{aligned}$$

一般地, 边界点 a, b 固定 $\rightarrow \delta y|_{x=a} = \delta y|_{x=b} = 0$

$\hookrightarrow$ 欧拉方程: $\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} = 0$ ①

* $\frac{dF}{dx} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} y' + \frac{\partial F}{\partial y'} y''$

$\rightarrow \frac{\partial F}{\partial x} = \frac{dF}{dx} - \frac{\partial F}{\partial y} y' - \frac{\partial F}{\partial y'} y''$

$= \frac{d}{dx} \left[F - y' \frac{\partial F}{\partial y'} \right] - y' \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right]$

$\hookrightarrow$ 欧拉方程: $\frac{\partial F}{\partial x} + \frac{d}{dx} \left(y' \frac{\partial F}{\partial y'} - F \right) = 0$ ②

当 F 不显含 x ($\frac{\partial F}{\partial x} = 0$)

$\Rightarrow y' \frac{\partial F}{\partial y'} - F = \text{const}$

泛函导数: δy 在 x_0 附近不为 0, y 与 $y+\delta y$ 围成面积 δG

若 $\delta \left[\frac{\delta J}{\delta \epsilon} \right] \Big|_{x_0} = \frac{\delta J}{\delta y} \Big|_{x_0} = \lim_{\delta \epsilon \rightarrow 0} \frac{\delta J}{\delta \epsilon}$

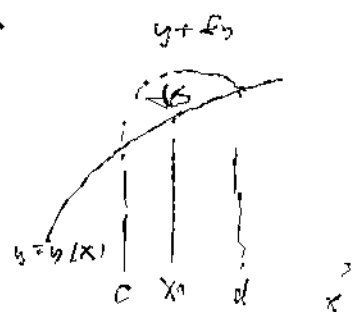
$\frac{\delta J}{\delta \epsilon} \Big|_{x_0} = \lim_{\delta \epsilon \rightarrow 0} \left\{ \frac{1}{\delta \epsilon} \int_c^d \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] \delta y \, dx \right\}$

$= \lim_{\delta \epsilon \rightarrow 0} \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right]_{x_0} \frac{1}{\delta \epsilon} \int_c^d \delta y \, dx$

$= \delta G$

$= \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right]_{x_0}$

$y = y(x)$ 是变分问题的解 $\Leftrightarrow$ 泛函 J 在任一点的泛函导数为 0



例：求解捷线问题

解：即求泛函 $J[y] = \int_0^1 \frac{\sqrt{1+y'^2}}{2y} dx$ 的极值。

$$\delta J[y] = 0 \Rightarrow \text{欧拉方程: } \frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \quad \text{此时 } F = \frac{\sqrt{1+y'^2}}{2y}$$

$$\text{代入得 } \left(-\frac{1}{2} \right) \frac{\sqrt{1+y'^2}}{2y^2} - \frac{d}{dx} \left(\frac{1}{2y} \cdot \frac{1}{2} \frac{2y'y'}{\sqrt{1+y'^2}} \right)$$

$$= -\frac{\sqrt{1+y'^2}}{2 \cdot 2y^2 \cdot y} - \frac{1}{2y} \left[\frac{y''}{y \sqrt{1+y'^2}} - y' \cdot \frac{1}{2} \frac{1}{y \sqrt{1+y'^2}} \right] \cdot (y' + y'^2 + 2yy'y'')$$

使用：

$$\frac{d}{dx} = y' \frac{d}{dy} \rightarrow \text{消去 } dx \text{ 项} = \text{所求}$$

假设 F 只是 x, y, y' 的函数时

$$\frac{d}{dx} = \frac{\partial}{\partial x} + \frac{\partial}{\partial y} \cdot y' + \frac{\partial}{\partial y'} \cdot y'' = \frac{\partial}{\partial x} + y' \left(\frac{\partial}{\partial y} + \frac{\partial}{\partial y'} \frac{dy'}{dx} \cdot \frac{dx}{dy} \right) = \frac{\partial}{\partial x} + y' \frac{d}{dy}$$

$$= -\frac{\sqrt{1+y'^2}}{2 \cdot 2y^2 \cdot y} - y' \frac{d}{dy} \left(\frac{y'}{2y^2 \sqrt{1+y'^2}} \right) = \frac{\partial}{\partial x} + y' \frac{d}{dy}$$

$$\Rightarrow \frac{\sqrt{1+y'^2}}{y^{3/2}} + 2y' \frac{d}{dy} \left(\frac{y'}{y \sqrt{1+y'^2}} \right) = 0$$

$$\Rightarrow \frac{\sqrt{1+y'^2}}{y^{3/2}} dy + 2y' \cdot \frac{(y \sqrt{1+y'^2} dy - y' \cdot 2 \sqrt{y \sqrt{1+y'^2}} dy + y'^2 dy + 2yy'y' dy)}{y^2 (1+y'^2)^{3/2}}$$

$$= \frac{\sqrt{1+y'^2}}{y^{3/2}} dy + y' \frac{2y \sqrt{1+y'^2} dy - y' (dy + y'^2 dy + 2yy'y' dy)}{y^{3/2} (1+y'^2)^{3/2}} = 0$$

$$\Rightarrow [1+y'^2]^2 dy + y' (2y dy - y' dy - y'^3 dy) = 0$$

$$[1+y'^2]^2 - y'^2 - y'^4 dy + 2yy' dy = 0$$

$$\Rightarrow \frac{dy}{2y} + \frac{y' dy'}{1+y'^2} = 0$$

积分：

$$\frac{1}{2} \ln(y) + \frac{1}{2} \ln(1+y'^2) + C_1 = 0$$

$$\text{得 } y' = [\exp(2C_1 - \ln y) - 1]^{1/2} = \sqrt{\frac{2C_1}{y} - 1}$$

$$\text{令 } dx = \frac{y}{2C_1 - y} dy$$

$$\text{其中 } \int \frac{\sqrt{y}}{2C_1 - y} dy \quad \text{令 } u = \sqrt{2C_1 - y} \quad \text{则 } y = 2C_1 - u^2 \quad dy = -2u du$$

$$= \int \frac{\sqrt{2C_1 - u^2}}{u} (-2u du) = (-2) \int \sqrt{2C_1 - u^2} du = (-2) \cdot \left(\frac{1}{2} C_1 \right) \int \sqrt{1 - \left(\frac{u}{\sqrt{C_1}}\right)^2} d\left(\frac{u}{\sqrt{C_1}}\right)$$

$$\text{令 } \frac{u}{\sqrt{C_1}} = \sin t$$

$$= -8C_1^2 \cdot \int \cos t (-\cos t) dt = 8C_1^2 \int \cos^2 t dt$$

$$\text{其中 } \int \cos^2 t dt = \int \frac{1}{2} [\cos(2t) + 1] dt = \frac{1}{2} t + \frac{1}{4} \sin(2t) + C$$

$$\begin{aligned} \int \frac{\sqrt{y}}{2C_1 - y} dy &= C_1^2 \left[4t + 2\sin(2t) + 8C \right] \\ &= C_1^2 \left[4 \arccos\left(\frac{\sqrt{2C_1 - y}}{\sqrt{C_1}}\right) + 4 \frac{\sqrt{2C_1 - y}}{\sqrt{C_1}} \cdot \sqrt{1 - \frac{2C_1 - y}{4C_1^2}} + 8C \right] \end{aligned}$$

$$= -\sqrt{2C_1 y - y^2} + C_1 \arccos \frac{C_1 - y}{C_1} + C_2$$

$$\text{令 } \theta = \arccos \frac{C_1 - y}{C_1}$$

$$\text{则 } x = C_1(\theta - \sin \theta) + C_2$$

$$y = C_1(1 - \cos \theta)$$

* 考虑到 F 不是全 x ($\frac{\partial F}{\partial x} = 0$)

可以使用欧拉方程:

$$y' \frac{\partial F}{\partial y'} - F = y' \cdot \frac{y'}{\sqrt{1+y'^2}} - \frac{\sqrt{1+y'^2}}{\sqrt{2yy'}} = -\frac{1}{\sqrt{2yy'}} \cdot \sqrt{1+y'^2} = \cos 2$$

$$\text{亦可解: } \begin{cases} x = \\ y = \end{cases}$$

变分情况

(1) 多个函数

$$J = \int_a^b F(x, y_1, y_2, \dots, y_s, y_1', y_2' \dots y_s') dx$$

$$\delta J = 0, \text{ 在固定的边界下 } (\delta y|_a = \delta y|_b = 0)$$

$$\hookrightarrow \frac{\partial F}{\partial y_\alpha} - \frac{d}{dx} \left(\frac{\partial F}{\partial y_\alpha'} \right) = 0 \quad \alpha = 1 \dots s$$

(2) 高阶导数

$$J = \int F(x, y, y' \dots y^{(n)}) dx$$

$$\delta J = 0$$

$$\hookrightarrow \frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \dots + (-1)^n \frac{d}{dx^n} \left(\frac{\partial F}{\partial y^{(n)}} \right) = 0$$

(3) 二元函数

$$J = \int F(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}) dx \quad \text{其中 } u = u(x, y)$$

$$\delta J = 0$$

$$\hookrightarrow \frac{\partial F}{\partial u} - \frac{\partial}{\partial x} \left[\frac{\partial F}{\partial \frac{\partial u}{\partial x}} \right] - \frac{\partial}{\partial y} \left[\frac{\partial F}{\partial \frac{\partial u}{\partial y}} \right] = 0$$

4. 约束条件下的变分问题

$$\text{泛函 } J = \int_a^b F(x, y_1, \dots, y_s, y_1', \dots, y_s') dx \text{ 取决于 } s \text{ 个 } y_\alpha(x)$$

$$\text{且有 } k \text{ 个约束条件: } \int_a^b G_j(x, y_1, \dots, y_s, y_1', \dots, y_s') dx = C_j$$

$$j = 1, \dots, k$$

此时原 s 个欧拉方程中有 k 个不独立

使用 Lagrange 乘子法.

对约束两侧取变分, 有

$$\delta \int_a^b G_j(x, y_1, \dots, y_s, y_1', \dots, y_s') dx = 0$$

$$\sum_{\alpha=1}^s \left[\frac{\partial F}{\partial y_{\alpha}} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'_{\alpha}} \right) \right] \delta y_{\alpha} dx = 0 \quad] = 1 - k$$

选取 k 个 $\lambda_j > 0$

$$\int_a^b \sum_{\alpha=1}^s \left[\frac{\partial F}{\partial y_{\alpha}} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'_{\alpha}} \right) + \sum_{j=1}^k \lambda_j \left[\frac{\partial G_j}{\partial y_{\alpha}} - \frac{d}{dx} \left(\frac{\partial G_j}{\partial y'_{\alpha}} \right) \right] \right] \delta y_{\alpha} dx = 0$$

Q 这样选取 λ_j 的各部分就满足 (?)

$$\rightarrow \frac{\partial}{\partial y_{\alpha}} \left(F + \sum_{j=1}^k \lambda_j G_j \right) - \frac{d}{dx} \frac{\partial}{\partial y'_{\alpha}} \left(F + \sum_{j=1}^k \lambda_j G_j \right) = 0$$

此时新的 s 个方程 + k 个约束

决定 s 个 y_{α} , k 个 λ_j

例: 设平面上封闭曲线长度为 l , 求其围成面积的最大值.

$$\text{解: 设 } y(x) = \begin{cases} x = x(z) & \text{封闭} \rightarrow x(0) = x(z_0) \\ y = y(z) & y(0) = y(z_0) \end{cases}$$

$$\text{约束: } l = \oint_C [(dx)^2 + (dy)^2]^{1/2} = \int_0^{z_0} [x'^2(z) + y'^2(z)]^{1/2} dz$$

$$\text{面积 } S = \iint_D dx dy$$

$$\text{Green 定理: } \oint_C (P dx + Q dy) = \oint_C \vec{A} \cdot d\vec{r}$$

$$\nabla \times \vec{A} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \times (A_x, A_y, A_z) = (0, 0, 1)$$

$$\rightarrow A_x = -\frac{1}{2}y, A_y = \frac{1}{2}x, A_z = 0$$

$$\text{因此 } S = \frac{1}{2} \oint_C (x dy - y dx) = \frac{1}{2} \int_0^{z_0} (x y' - y x') dz$$

$$\text{Lagrange 函数 } L = \left[\frac{1}{2} (x y' - y x') + \lambda (x^2 + y^2)^{1/2} \right]$$

$$\text{欧拉方程: } \begin{cases} \frac{\partial L}{\partial x} - \frac{d}{dz} \left(\frac{\partial L}{\partial x'} \right) = 0 \\ \frac{\partial L}{\partial y} - \frac{d}{dz} \left(\frac{\partial L}{\partial y'} \right) = 0 \end{cases}$$

$$y' = \frac{dy}{dx}$$

$$\begin{cases} y' - \frac{d}{dx} \left(\frac{\lambda x'}{\sqrt{x'^2 + y'^2}} \right) = 0 \\ x' + \frac{d}{dx} \left(\frac{\lambda y'}{\sqrt{x'^2 + y'^2}} \right) = 0 \end{cases}$$

另法

$$\begin{cases} y - \frac{\lambda x'}{\sqrt{x'^2 + y'^2}} = C_1 \\ x + \frac{\lambda y'}{\sqrt{x'^2 + y'^2}} = C_2 \end{cases} \Rightarrow \begin{cases} y - C_1 = \frac{\lambda x'}{\sqrt{x'^2 + y'^2}} & ① \\ x - C_2 = \frac{\lambda y'}{\sqrt{x'^2 + y'^2}} & ② \end{cases}$$

$$①^2 + ②^2 \quad (y - C_1)^2 + (x - C_2)^2 = \lambda^2$$

$$\text{代入约束 } L = \oint_C (ax)^2 + (ay)^2 = 2\pi\lambda \Rightarrow \lambda = \frac{L}{2\pi}$$

§ 8.2 哈密顿原理

1. 拉格朗日方程 $\sim$ 欧拉方程

$$\delta \int_{t_1}^{t_2} L(t, q, \dot{q}) dt = 0$$

定义:

哈密顿作用量 $S = \int L dt \Rightarrow \delta S = 0$ S 取极值 哈密顿原理

位形空间 q_1, q_2 为坐标轴
位形世界 q_1, q_2, t

达朗贝尔原理 $\rightarrow$ 哈密顿原理:

$$\sum_i (\vec{F}_i - m_i \ddot{\vec{r}}_i) \cdot \delta \vec{r}_i = 0$$

$$m_i \ddot{\vec{r}}_i \cdot \delta \vec{r}_i = m_i \frac{d}{dt} \dot{\vec{r}}_i \cdot \delta \vec{r}_i$$

分部积分

$$= \frac{d}{dt} (m_i \dot{\vec{r}}_i \cdot \delta \vec{r}_i) - m_i \dot{\vec{r}}_i \cdot \delta \dot{\vec{r}}_i$$

$$= \frac{d}{dt} (m_i \dot{\vec{r}}_i \cdot \delta \vec{r}_i) - \delta \left(\frac{1}{2} m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i \right)$$

$$= \frac{d}{dt} (m_i \dot{\vec{r}}_i \cdot \delta \vec{r}_i) - \delta T_i$$

可积?

两边 $\int_{t_1}^{t_2}$

$$\int \delta T_i dt = \delta \int T_i dt$$

$$\int_{t_1}^{t_2} m_i \ddot{\vec{r}}_i \cdot \delta \vec{r}_i = (m_i \dot{\vec{r}}_i \cdot \delta \vec{r}_i) \Big|_{t_1}^{t_2} - \delta \int_{t_1}^{t_2} T_i dt$$

初、末位形给定 $\rightarrow \delta \vec{r}_i \Big|_{t_1} = \delta \vec{r}_i \Big|_{t_2} = 0$

$$\int_{t_1}^{t_2} \vec{F}_i \cdot \delta \vec{r}_i = \int_{t_1}^{t_2} \delta W_i = \delta \int_{t_1}^{t_2} W_i dt$$

$$\Rightarrow \delta \int_{t_1}^{t_2} \sum_i (W_i + T_i) dt = 0$$

$$\text{即 } \delta \int_{t_1}^{t_2} (T + W) dt = 0$$

1. 已知 $\delta W = \sum \bar{Q}_d dq_d$, $\bar{Q}_d = -\delta \bar{q}_d$

$$\Rightarrow \delta \int_{t_1}^{t_2} (\bar{T} - \bar{V}) dt = \delta \int_{t_1}^{t_2} L dt = 0$$

2. 相空间中的哈密顿原理

$$\delta \int_{t_1}^{t_2} [\sum \bar{p}_d \dot{q}_d - H(p, q, t)] dt = 0$$

$$\int_{t_1}^{t_2} [\sum (\bar{p}_d \delta \dot{q}_d + \dot{q}_d \delta \bar{p}_d - \frac{\partial H}{\partial \bar{p}_d} \delta \bar{p}_d - \frac{\partial H}{\partial \bar{q}_d} \delta \bar{q}_d)] dt = 0$$

其中: $\int_{t_1}^{t_2} \sum \bar{p}_d \delta \dot{q}_d dt$

$$= \int_{t_1}^{t_2} \frac{d}{dt} [\bar{p}_d \delta q_d] - \bar{p}_d \delta q_d dt$$

$$= \bar{p}_d \delta q_d \Big|_{t_1}^{t_2} - \int \bar{p}_d \delta q_d dt$$

初、末位置固定: $\delta q_d \Big|_{t=t_1} = \delta q_d \Big|_{t=t_2} = 0$

$$= - \int_{t_1}^{t_2} \bar{p}_d \delta q_d dt$$

$$= \int_{t_1}^{t_2} \sum \left[\dot{q}_d - \frac{\partial H}{\partial \bar{p}_d} \right] \delta \bar{p}_d - \left[\frac{\partial H}{\partial \bar{q}_d} + \bar{p}_d \right] \delta \bar{q}_d dt = 0$$

$$\Rightarrow \begin{cases} \dot{q}_d = \frac{\partial H}{\partial \bar{p}_d} \\ \bar{p}_d = - \frac{\partial H}{\partial \bar{q}_d} \end{cases}$$

* 在被积函数中加上某个函数 u 对时间全导数.

$$\delta \int_{t_1}^{t_2} \frac{du}{dt} dt = \delta u \Big|_{t_1}^{t_2}$$

在相空间(位形)空间中, 初、末位置固定

$$\delta u \Big|_{t=t_1} = \delta u \Big|_{t=t_2} = 0 \quad \checkmark \quad (7)$$

$\Rightarrow$ 不影响动力学方程

3. 位形世界中的哈密顿原理:

t 与 $q_1, \dots, q_s$ 平等

$$q_d = q_d(t) \quad t = t(t) \quad \dot{q}_d = \frac{\partial q_d}{\partial t} = \frac{\frac{\partial q_d}{\partial t}}{\frac{\partial t}{\partial t}} = \frac{\dot{q}_d}{1}$$

$$S = \int L(t, q, \dot{q}, t) dt = 0$$

定义：修改拉格朗日函数 Λ

$$\Lambda = L(q, t, \frac{q'}{c}, t')$$

$$\delta \int \Lambda dt = 0$$

$$\text{欧拉方程: } \frac{d}{dt} \left(\frac{\partial \Lambda}{\partial \dot{q}_\alpha} \right) - \frac{\partial \Lambda}{\partial q_\alpha} = 0 \quad (1)$$

$$\frac{d}{dt} \frac{\partial \Lambda}{\partial t'} - \frac{\partial \Lambda}{\partial t} = 0 \quad (2)$$

$$\text{其中 } (1) \quad \frac{\partial \Lambda}{\partial q_\alpha} = \frac{\partial (L t')}{\partial q_\alpha} = \frac{\partial L}{\partial q_\alpha} t'$$

$$= \frac{\partial L}{\partial q_\alpha} \left(\frac{\partial q_\alpha}{\partial q_\alpha} t' \right)$$

$$= \frac{\partial L}{\partial q_\alpha} \left[\frac{\partial}{\partial q_\alpha} \left(\frac{q_\alpha}{t} \right) \right] t' = \frac{\partial L}{\partial q_\alpha}$$

$$\frac{\partial \Lambda}{\partial q_\alpha} = \frac{\partial (L t')}{\partial q_\alpha} = \frac{\partial L}{\partial q_\alpha} t' = \frac{\partial L}{\partial q_\alpha} \frac{\partial t}{\partial t'}$$

$$\frac{d}{dt} = \frac{d}{dt'} \left(\frac{\partial t}{\partial t'} \right) \Rightarrow \frac{\partial t}{\partial t'} = \frac{\frac{dt}{dt'}}{\frac{dt}{dt'}}$$

$$(2) \quad \frac{\partial \Lambda}{\partial t'} = \frac{\partial (L t')}{\partial t'} = L + \frac{\partial L}{\partial t'} t'$$

$$= L + \sum_{\alpha} \frac{\partial L}{\partial q_\alpha} \frac{\partial q_\alpha}{\partial t'} t' = L + \sum_{\alpha} \left[\frac{\partial L}{\partial q_\alpha} \frac{\partial}{\partial t'} \left(\frac{q_\alpha}{t'} \right) \right] t'$$

$$= L - \sum_{\alpha} \frac{\partial L}{\partial q_\alpha} \frac{q_\alpha}{t'^2} t' = L - \sum_{\alpha} \frac{\partial L}{\partial q_\alpha} q_\alpha$$

$$= L - \sum_{\alpha} p_\alpha \dot{q}_\alpha = -H$$

$$\frac{\partial \Lambda}{\partial t} = \frac{\partial (L t')}{\partial t} = t' \frac{\partial L}{\partial t} + L \frac{\partial t'}{\partial t}$$

$$\Rightarrow (1) \quad \frac{d}{dt} \left(\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_\alpha} \right) - \frac{\partial L}{\partial q_\alpha} \right) = 0$$

$$\textcircled{2} \quad -\frac{dH}{dt} - \frac{\partial L}{\partial t} = -\frac{dH}{dt} - \frac{\partial(Lt')}{\partial t} = -\frac{dH}{dt} - t' \frac{\partial L}{\partial t} \\ = -\frac{dH}{dt} - \frac{dt}{dt} \frac{\partial L}{\partial t} = -\frac{dt}{dt} \left(\frac{dH}{dt} + \frac{\partial L}{\partial t} \right) = 0$$

137: 弹性摆

k, l_0, m



$$\textcircled{1} \quad L = T - V = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) - [-m g r \cos \theta + \frac{1}{2} k (r - l_0)^2]$$

$$\delta \int L dt = \delta \int \left[\frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) + m g r \cos \theta - \frac{1}{2} k (r - l_0)^2 \right] dt$$

$$= \int [m \dot{r} \delta r + m r \dot{\theta}^2 \delta r + m r^2 \dot{\theta} \delta \theta + m g \cos \theta \delta r - m g r \sin \theta \delta \theta - k(r - l_0) \delta r] dt$$

对含 δr , $\delta \theta$ 项进行分部积分

$$r \delta \dot{r} = \frac{d}{dt} (r \delta r) - \dot{r} \delta r$$

$$r^2 \dot{\theta} \delta \dot{\theta} = \frac{d}{dt} (r^2 \dot{\theta} \delta \theta) - 2 r \dot{r} \dot{\theta} \delta \theta - r^2 \ddot{\theta} \delta \theta$$

$$= \int [-m \ddot{r} + m r \dot{\theta}^2 + m g \cos \theta - k(r - l_0)] \delta r + [-2 m r \dot{r} \dot{\theta} - m r^2 \ddot{\theta} - m g r \sin \theta] \delta \theta dt \\ + m r \delta r \Big|_{t_1}^{t_2} + m r^2 \dot{\theta} \delta \theta \Big|_{t_1}^{t_2} = 0 \quad \delta r|_{t=t_1} = \delta r|_{t=t_2} = 0, \delta \theta|_{t=t_1} = \delta \theta|_{t=t_2} = 0$$

$$\Rightarrow \begin{cases} m \ddot{r} = m r \dot{\theta}^2 + m g \cos \theta - k(r - l_0) \\ m r \ddot{\theta} = -m g \sin \theta - 2 m \dot{r} \dot{\theta} \end{cases}$$

$$\textcircled{2} \quad p_r = \frac{\partial L}{\partial \dot{r}} = m \dot{r} \quad p_\theta = \frac{\partial L}{\partial \dot{\theta}} = m r^2 \dot{\theta}$$

$$H = \dot{r} p_r + \dot{\theta} p_\theta - L = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} + \frac{1}{2} k (r - l_0)^2 - m g r \cos \theta$$

$$\delta \int (\dot{r} p_r + \dot{\theta} p_\theta - H) dt = 0$$

↓

$$\begin{cases} -\dot{p}_r + \frac{p_\theta^2}{mr^3} - k(r - l_0) + m g \cos \theta \\ \dot{r} - \frac{p_r}{m} = 0 \\ -\dot{p}_\theta + \frac{p_r p_\theta}{mr^2} - m g \sin \theta = 0 \end{cases}$$

$$-10 = \ln 9 \quad \Rightarrow \neq 0$$

$$0 - \frac{P_0}{n_1 n_2} = 0$$

§3 最小作用量原理

1. 若 q_i 是可选坐标

$$\text{则 } p_i = \frac{\partial L}{\partial \dot{q}_i} = \text{const.}$$

↓ 可以解出

$$q_i = q_i(q_2 - q_5, p_i, q_2 - q_5, t)$$

⇒ 在 $\delta \int L dt = 0$ 中减少了自由度:

X 此时 δq_i 不再独立, $\delta q_i|_{t=t_1}, \delta q_i|_{t=t_2}$ 不一定为 0

$$\delta \int L dt = \int \sum \left(\frac{\partial L}{\partial q_\alpha} \delta q_\alpha + \frac{\partial L}{\partial \dot{q}_\alpha} \delta \dot{q}_\alpha \right) dt$$

分部积分

$$= \int \sum \left[\frac{\partial L}{\partial q_\alpha} \delta q_\alpha + \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_\alpha} \delta q_\alpha \right) - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_\alpha} \right) \delta q_\alpha \right] dt$$

$$= \frac{\partial L}{\partial \dot{q}_\alpha} \delta q_\alpha \Big|_{t_1}^{t_2} = p_\alpha \delta q_\alpha \Big|_{t_1}^{t_2}$$

$$\Rightarrow \delta \int (L - p_i \dot{q}_i) dt = 0$$

修改后的拉格朗日函数

2.

当 L 不显含 t

$$p_t = \frac{\partial L}{\partial t} = -H$$

$$1 - p_t t' = (L - p_t \dot{q}) t' = (L + H) t' = \sum p_\alpha \dot{q}_\alpha t'$$

定义, 雅可比作用量

$$W = \int \sum p_\alpha \dot{q}_\alpha t' dt$$

$$\Delta W = \delta \int \sum p_\alpha \dot{q}_\alpha t' dt = 0$$

↓ 不等时变分 (全变分)

$$\Delta q_\alpha = \delta q_\alpha + \dot{q}_\alpha \delta t$$

$$\text{欧拉定理 } \sum p_\alpha \dot{q}_\alpha = \sum \frac{\partial T}{\partial \dot{q}_\alpha} \dot{q}_\alpha = 2T$$

$$\rightarrow \oint 2T dt = 0, \quad \oint 2T dt = 0$$

$$2T dt = m v^2 \frac{ds}{v} = m v ds = \sqrt{2mT} ds = \sqrt{2m(E - V)} ds$$

$$\rightarrow \oint m v ds = 0 \quad \sim \text{量子原理}$$

$$\text{定义位形空间中的弧长 } (ds)^2 = \sum_\alpha \sum_\beta m_{\alpha\beta} dq_\alpha dq_\beta$$